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Neural Demodulation of Hardware-Captured PSK and Star-QAM Waveforms Under a Provable Memorisation Bound

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ABSTRACT

Learned receivers are often evaluated on data whose transmitted sequence is short and periodic, which lets an expressive model predict a symbol from its neighbours instead of demodulating the waveform. We exclude this failure mode by construction. Because the transmitted sequence has linear order six, any six consecutive bits determine the seventh; restricting the receiver's input to at most five bits therefore makes prediction-from-context mathematically impossible, not merely unlikely. We test this bound on a hardware dataset of 154,628 labelled symbols, captured from five physically distinct trainer boards across BPSK, QPSK and star 8-QAM, with ground truth measured directly from the transmitted waveform rather than trusted from the transmitter's configuration. A 21,000-parameter convolutional receiver trained under this bound matches the symbol error rate of a blind matched-filter receiver on all three modulations, despite receiving no carrier frequency, pulse shape or constellation information, and its errors concentrate on the nearest constellation neighbours— evidence that it has recovered the true signal geometry rather than memorised labels. Holding out an entire transmitted word costs almost nothing in accuracy, consistent with what the bound predicts; holding out a device costs measurably more, pointing to cross-device generalisation as the harder open problem for this class of receiver. We also report two negative results: a constellation-aware loss term does not improve accuracy, and a widely proposed carrier-sensitivity disadvantage for classical receivers does not hold up under direct measurement.

INDEX TERMS Neural receiver, demodulation, deep learning, physical layer, hardware dataset, maximal-length sequence, memorisation, symbol error rate, quadrature amplitude modulation.

I. INTRODUCTION

LEARNING-based receivers recover digital modulation from sampled waveforms without an explicitly specified signal model [1]–[3]. A recurring difficulty in evaluating such systems is that the transmitted data is frequently a short pseudorandom sequence repeated many times. When that is so, a model with sufficient temporal context can predict the next symbol from preceding symbols alone, without extracting any information from the waveform. The reported error rate then measures sequence memorisation rather than demodulation, and held-out evaluation does not detect this if the same short sequence appears in both partitions.

This paper takes the position that the property should be excluded structurally rather than argued empirically. The transmitters studied here emit a maximal-length sequence of linear order six. For such a sequence, any six consecutive bits determine the seventh exactly, while any five do not. A receiver whose input contains at most five bits therefore cannot perform prediction-from-context, irrespective of its capacity, its training procedure, or the quantity of data available to it. We enforce this by limiting the receiver's input window, so that the bound is a property of the data presented to the network rather than of its architecture, and can be verified by inspection.

The contributions are as follows.

1) A hardware dataset of 43 dual-channel oscilloscope captures from five physically distinct trainer boards, spanning BPSK, QPSK and a two-ring star 8-QAM

constellation, with a per-record verification protocol and full provenance (Section III).
2) A ground-truth procedure based on coherent averaging of the transmitted waveform, with two independent confirmations that the recovered labels are correct (Section IV).
3) A receptive-field bound that makes sequence memorisation impossible, stated as a proof and confirmed empirically by the absence of any gap between a random split and a leave-one-pattern-out split (Sections V-A and VI-B).
4) A characterisation of the boards' own demodulators, including a lock threshold reproducing across all five units, and evidence that self-consistency is an invalid criterion for demodulator lock (Section VI-E).
5) Two negative results, reported because they bear directly on claims made elsewhere in this literature: a constellation-geometry loss term does not improve error rate, and a blind classical receiver's dependence on carrier knowledge does not constitute a practical disadvantage at any observation length tested (Sections VI-D and VIII).

We make no claim of improved error-rate performance. The learned receiver attains parity with a blind classical receiver; the contribution is that it does so without being supplied the carrier frequency, the pulse shape, the constellation geometry or the decision boundaries, and that it provably cannot have obtained its result by memorisation.

II. RELATED WORK

A. LEARNED PHYSICAL-LAYER PROCESSING

Learned physical-layer processing has been demonstrated for end-to-end autoencoder transceivers [1], for systems trained over the air [2], and for joint channel estimation and detection [3]. Modulation classification from sampled in-phase and quadrature data has received particular attention, together with the synthetic datasets used to benchmark it [4]. Broad reviews of the area are given in [5]–[7], and of distribution shift specifically in [8].

The useful axis for comparison is what each system is *given*. Table 1 records, for twenty-three recent learned receivers and demodulators, whether the carrier frequency, the symbol timing, the pulse shape and the constellation are supplied rather than recovered, together with the domain of the input and the origin of the evaluation data.

The distribution is uneven. Ten operate on baseband in-phase and quadrature samples and eight on frequency-domain subcarriers; four take symbols already at the output of a matched filter, and one operates on passband samples. Seventeen are given the carrier frequency and sixteen the symbol timing. Twenty are given the constellation, and nineteen fix a known pulse-shape family without presenting it to the network. Twenty-one are evaluated on simulated waveforms; the two exceptions use hardware captures for validation only.

A minority withhold more. Zhang *et al.* demodulate BPSK with a sliding-window one-dimensional convolutional network on raw passband samples, detecting phase-shift locations without explicit carrier synchronisation or symbol-period grouping [17]; that system is the closest to ours in input domain, and differs in being evaluated on simulated waveforms and in being given the constellation. Xiao *et al.* take raw baseband samples and compensate time and frequency offsets blindly while jointly predicting the modulation type and the symbol index [12]. Dakic *et al.* predict LoRa symbols directly from baseband samples and report robustness to emulated carrier and time offsets without explicit synchronisation modules [18]. Zhu *et al.* recover bit sequences from oversampled baseband sequences while handling symbol timing offsets blindly through a shifted binary cross-entropy loss [10]. Hanna *et al.* estimate carrier offset, timing and symbol rate blind and then decode with a minimum-distance receiver against the classified modulation's constellation [11], and are one of the two entries whose evaluation involves hardware.

The larger group operates in the frequency domain with synchronisation already established. DeepRx performs channel estimation, equalisation and soft demapping jointly and emits bit log-likelihood ratios for 5G [27], and its MIMO extension adds either a combining-based or a fully learned transformation layer [28]. Cammerer *et al.* train a bit-wise autoencoder that learns geometric constellation shaping and bit labelling jointly with iterative demapping and decoding [29], making it one of the three tabulated systems not given a constellation. A further group learns the demapping alone, from symbols already at the matched-filter output [19]–[22].

These systems are evaluated on held-out partitions. Twelve of the twenty-three report a random held-out split; the remaining eleven do not state one. None reports a partition grouped by recording or by device, and none constrains the receiver's input context relative to the complexity of the transmitted sequence, which is the property Section V-A establishes structurally rather than argues empirically.

B. EVALUATION VALIDITY AND MEMORISATION

This paper's premise is that held-out evaluation is insufficient when the transmitted sequence is short and periodic. Related concerns are long-standing in machine learning generally—train/test contamination, and models that exploit dataset artefacts rather than the intended signal—but the specific failure considered here is narrower: the test sequence itself is predictable from its own past, so a model with sufficient context can reach a low error rate without demodulating.

Work in radio-frequency sensing has begun to quantify the effect. Shulman shows that segment-level cross-validation causes severe data leakage and optimism in drone identification benchmarks, and that regrouping the evaluation by recording collapses the inflated accuracies [30]; among the studies surveyed here it is the only one that reports a recording-grouped split. The same failure is documented outside wireless: Brookshire *et al.* show that dividing recordings into segments without grouping by subject produces severe leakage and heavily inflated performance in translational electroencephalography studies [31].

TABLE 1. What recent learned receivers and demodulators are given. *family* denotes a fixed known pulse-shape family that is not itself presented to the network; *blind* denotes symbol timing recovered by the system; *partial* denotes a residual offset estimated from the signal; *n.s.* denotes a property the paper does not state, which is not the same as a property it does not use. *Domains*: passband, baseband in-phase and quadrature, post-matched-filter symbols, frequency domain. *Data*: simulated, hardware used for validation only, hardware as the primary source. Surveys without a system are cited in the text and not tabulated.

System	Carrier	Timing	Pulse	Const.	Domain	Data
Zhu *et al.* [9]	yes	blind	family	no	baseband	sim.
Zhu *et al.* [10]	yes	blind	family	yes	baseband	sim.
Hanna *et al.* [11]	partial	blind	family	yes	baseband	hw-valid.
Xiao *et al.* [12]	no	blind	family	no	baseband	sim.
Zhu *et al.* [13]	yes	blind	family	yes	baseband	sim.
Chia *et al.* [14]	yes	yes	family	yes	baseband	sim.
Tonix-Gleason *et al.* [15]	yes	yes	family	yes	freq.	sim.
Liu *et al.* [16]	yes	yes	family	yes	freq.	sim.
Zhang *et al.* [17]	no	blind	family	yes	passband	sim.
Dakic *et al.* [18]	no	blind	family	yes	baseband	sim.
Zheng *et al.* [19]	no	yes	yes	yes	symbols	sim.
Allwin *et al.* [20]	yes	yes	yes	yes	symbols	sim.
Gansekoele *et al.* [21]	yes	yes	yes	yes	symbols	sim.
Polvani *et al.* [22]	yes	yes	yes	yes	symbols	sim.
Wang *et al.* [23]	yes	yes	family	yes	baseband	sim.
Chen *et al.* [24]	yes	yes	family	yes	freq.	sim.
Gorday *et al.* [25]	no	yes	family	yes	baseband	sim.
Fischer *et al.* [26]	yes	yes	family	yes	freq.	sim.
Honkala *et al.* [27]	yes	yes	family	yes	freq.	sim.
Korpi *et al.* [28]	yes	yes	family	yes	freq.	sim.
Cammerer *et al.* [29]	yes	yes	family	no	freq.	hw-valid.
Hoydis *et al.* [5]	yes	yes	family	yes	freq.	sim.
Sathyanarayanan *et al.* [4]	yes	yes	family	yes	baseband	sim.
This work	no	yes	family	no	passband	hw-primary

The underlying concern is long-standing in machine learning generally. Carlini *et al.* give a quantitative framework for testing whether a sequence model has unintentionally memorised rare training sequences [32], and Arpit *et al.* show that gradient-based optimisation learns simple patterns before it memorises noise [33]. These results establish that the partition matters, and that memorisation is measurable after training. None of them bounds what a receiver can memorise before training begins. We are not aware of prior work in learned communications that constrains receiver context below the linear complexity of the test sequence, and that absence is part of the motivation for Section V-A.

C. HARDWARE-CAPTURED RF DATASETS

Public RF datasets are commonly synthesised, or captured from a small number of transmitters with ground truth asserted from the transmitter's configuration rather than measured from the emitted waveform. The distinction matters here: Section III shows a case in which the configured data word is *not* a reliable statement of the transmitted symbol sequence, and Section IV obtains labels by measurement instead.

Table 2 lists twelve public radio-frequency datasets against three axes: the number of physical devices, the granularity of the labels, and whether the data are captured from hardware or simulated. Nine are captured primarily from hardware, one uses hardware for validation only, and two are simulated.

The axis that matters here is label granularity. Seven of the twelve label at device level, three at modulation-class level, and two do not state a granularity. None labels individual symbols. Hanna *et al.* release the largest, ten million packets from 174 transmitters recorded across 41 receivers over a month [34]; Elmaghbub *et al.* capture 25 LoRa transceivers across indoor, outdoor and wired testbeds [37]; Zahid *et al.* capture 27 radios [38]; Morin *et al.* twenty [39]; Kashani *et al.* thirteen [42]; and Jagannath *et al.* ten commercial chipsets [43]. Our five units are fewer than any of these, and far fewer than the 174 of [34].

The distinction we draw is therefore not scale but what a label denotes. The datasets above answer which device transmitted a signal, or which modulation it carries. A symbol-

TABLE 2. Public radio-frequency datasets. *n.s.* denotes a quantity the dataset does not state. The device counts are as reported; ours is the smallest in the table.

Dataset	Devices	Label	Source
Hanna *et al.* [34]	174	device	hw-primary
Vrieze *et al.* [35]	n.s.	mod.-class	hw-primary
Al-Shawabka *et al.* [36]	n.s.	device	hw-valid.
Elmaghbub *et al.* [37]	25	device	hw-primary
Zahid *et al.* [38]	27	device	hw-primary
Morin *et al.* [39]	20	device	hw-primary
Utrilla [40]	n.s.	mod.-class	hw-primary
Allahham *et al.* [41]	3	mod.-class	hw-primary
Kashani *et al.* [42]	13	device	hw-primary
Jagannath *et al.* [43]	10	device	hw-primary
Jaensch *et al.* [44]	n.s.	n.s.	sim.
Yapar *et al.* [45]	n.s.	n.s.	sim.
This work	5	symbol	hw-primary

TABLE 3. Acquisition and signal parameters. All values were measured from the captured records rather than taken from the equipment specification.

Parameter	Value
Sample rate	87,260 Sa/s
Samples per record	523,600
Record duration	6.0004 s
Channel 1	TP-20, modulated output
Channel 2	TP-9, recovered data
Trigger	external, from TP-1
Carrier	1199.91 Hz
Symbol rate	599.955 baud
Carrier cycles per symbol	2 (exact)
Samples per symbol	145.44
BPSK data source	on-board 63-bit m-sequence
QPSK, 8-QAM data source	24-bit configurable words

level label answers which constellation point was transmitted at a given instant, which is what a demodulator must predict, and which Section IV obtains by measurement from the waveform rather than by assertion from the transmitter's configuration.

D. CLASSICAL RECEIVER THEORY

Classical treatments of the receiver functions replaced by learned systems carrier recovery, timing recovery, matched filtering and detection are given in [46] and [47]. The timing-error detector referenced in Section VIII is due to [48]. Properties of maximal-length sequences, including the linear recursion exploited in Section V-A, are standard and are set out in [49].

III. EXPERIMENTAL SETUP AND DATASET

A. APPARATUS

Signals were captured from MCM31/EV communication trainer boards using a two-channel digital oscilloscope, shown in Fig. 1. Channel 1 observes TP-20, the modulated output, and channel 2 observes TP-9, the board's own recovered data. The two are acquired simultaneously so that network input and hardware baseline originate from a single record. Acquisition is externally triggered from the board's symbol clock at TP-1.

Acquisition parameters were fixed for every record and are given in Table 3. The time base was chosen so that each record contains at least 57 repetitions of the longest data word, which is what makes the coherent averaging of Section IV effective. A shorter time base was evaluated and rejected: it yields too few repetitions for a stable signal-to-noise estimate.

The carrier is coherent with the symbol clock at exactly two cycles per symbol. This is the property that permits coherent averaging: the passband waveform repeats exactly at the word period, so averaging over repetitions suppresses noise without distorting the signal.

TABLE 4. Per-record verification battery. Every record in the corpus passes every check except where noted in Section III-D.

Check No.	Check	Pass condition
1	Channel count	2 channels present
2	Sample count	520,000–526,000
3	Duration	5.99–6.01 s
4	Sample rate	87,000–87,500 Sa/s
5	Quantisation levels	≥ 90 distinct
6	Carrier (4th power)	1199.5–1200.5 Hz
7	Word-period repetition	$r \geq 0.95$
8	Coherent-template SNR	≥ 15 dB
9a	Same word, two records	similarity ≥ 0.99
9b	Constellation fingerprint	matches the expected word
9c	Different words	similarity < 0.99
10	TP-9 agreement	logged, not gating

B. UNITS UNDER TEST

Ten units were sourced. Five were functional; five were excluded following individual inspection, in each case because the on-board data generator had failed, either in the sequence-load latch or in the pseudorandom generator itself. On such a unit the modulator receives no data and TP-20 carries an unmodulated carrier: the coherent template contains all of its energy in the carrier bin and none in the sidebands, and no usable waveform exists to capture. All results below therefore rest on five boards, and leave-one-board-out is a five-fold evaluation.

C. PER-RECORD VERIFICATION

Every record was subjected to the battery of checks in Table 4 before admission to the corpus. The battery is not decorative: it rejected records during the campaign and each rejection corresponded to a real configuration fault.

Elaborating check No.(9a–9c) from Table 4, we see that check No. (9a) compares two records of the same configured word and detects a failure to reload the transmitter between captures. Check No. (9b) compares the multiset of per-constellation-point occupancy counts against the value predicted by the configured word; it is invariant to constellation rotation and to the mapping between transmitter configuration and transmitted symbol, and therefore identifies the loaded word without assuming that mapping. Check No. (9c) is the converse and is required for held-out words, each of which is captured only once: two *different* configured words must not produce the same waveform. Neither (9a) nor (9b) detects a stale word in that case, because words designed to cover the full constellation share a fingerprint by construction.

Check No. (9c) must be evaluated on the *signed* similarity. Two words that are bitwise complements produce waveforms that are exact negations of one another, and an unsigned similarity reports these as identical when they are merely antipodal.

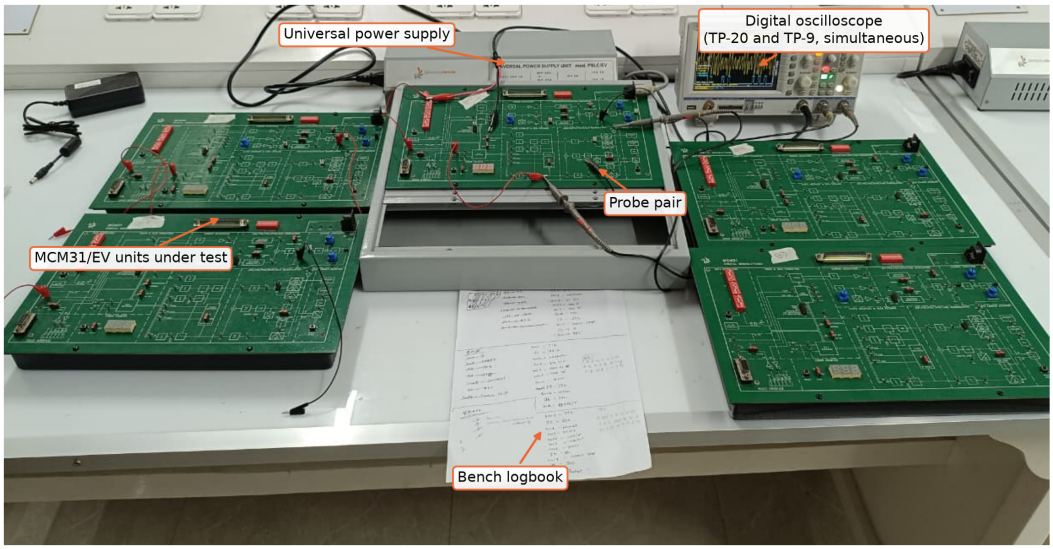


FIGURE 1. **Measurement bench. The five MCM31/EV trainer units, labelled Board-1 to Board-5, correspond to boards B1–B5 in Table 9 and Fig. 8. One unit is under test at a time, powered from a shared supply, with the oscilloscope probing TP-20 and TP-9 simultaneously and triggered externally from TP-1.**

TABLE 5. **Dataset composition. *Words* counts distinct configured bit patterns; *independent* accounts for the antipodal equivalence described in the text.**

Modulation	Records	Symbols	Words	Independent
BPSK	16	57,536	2	2
QPSK	12	43,152	3	3
8-QAM	15	53,940	6	5
Total	43	154,628		

D. CORPUS

The corpus comprises 43 records: fifteen from board B1 (seven main-campaign records and eight additional data words held out for evaluation) and seven from each of B2–B5. Segmentation at the symbol rate yields 154,628 labelled symbols, distributed as in Table 5. Coherent-template SNR across the accepted corpus is 22.7–43.2 dB, median 30.2 dB.

The distinction between *words* and *independent patterns* for 8-QAM is material. Two of the six words are bitwise complements of one another, and on this hardware complementary tribits map to antipodal constellation points: their coherent templates correlate at -0.998. After per-record amplitude normalisation the two carry identical information, and we therefore treat them as a single pattern throughout. Counting them separately would bias the leave-one-pattern-out result optimistically.

Four of the six 8-QAM words deliver six of the eight constellation points rather than the eight their bit patterns specify. The discriminator is a property computable from the bit pattern alone: a word is *framing robust* if all eight tribits survive every cyclic bit rotation. The two robust words deliver eight points; the four fragile words deliver six. This is consistent with the transmitter grouping bits into tribits from a fixed offset that does not coincide with the configuration order. It does not affect the labels, which are measured from the waveform, but it does mean the configured word is not a reliable statement of the transmitted symbol sequence which is precisely why the ground-truth procedure of Section IV does not use it.

E. MEASURED CONSTELLATIONS

Fig. 2 shows every symbol of every record, pooled across the five boards. The 8-QAM constellation is a four-phase, two-amplitude star of measured radius ratio 1.89 (range 1.867–1.914 over fifteen records). It is not a rectangular 16-QAM grid, and a loss function assuming rectangular geometry is inconsistent with this hardware.

IV. GROUND TRUTH

Labels are derived from the transmitted waveform, not from the transmitter's configuration. Fig. 3 shows the two processing paths and their separation.

A. COHERENT AVERAGING

Let $x[n]$ be the captured passband waveform and P the word period in samples, which is not in general an integer. The record is folded into K segments resampled onto a common grid of $M = \lfloor P \rceil$ points,

$$x_k[m;P] = x\big(kP + m\tfrac{P}{M}\big), \quad \begin{matrix} k = 0,\dots,K-1, \\ m = 0,\dots,M-1, \end{matrix} \tag{1}$$

and the template is their mean,

$$t[m;P] = \frac{1}{K}\sum_{k=0}^{K-1} x_k[m;P]. \tag{2}$$

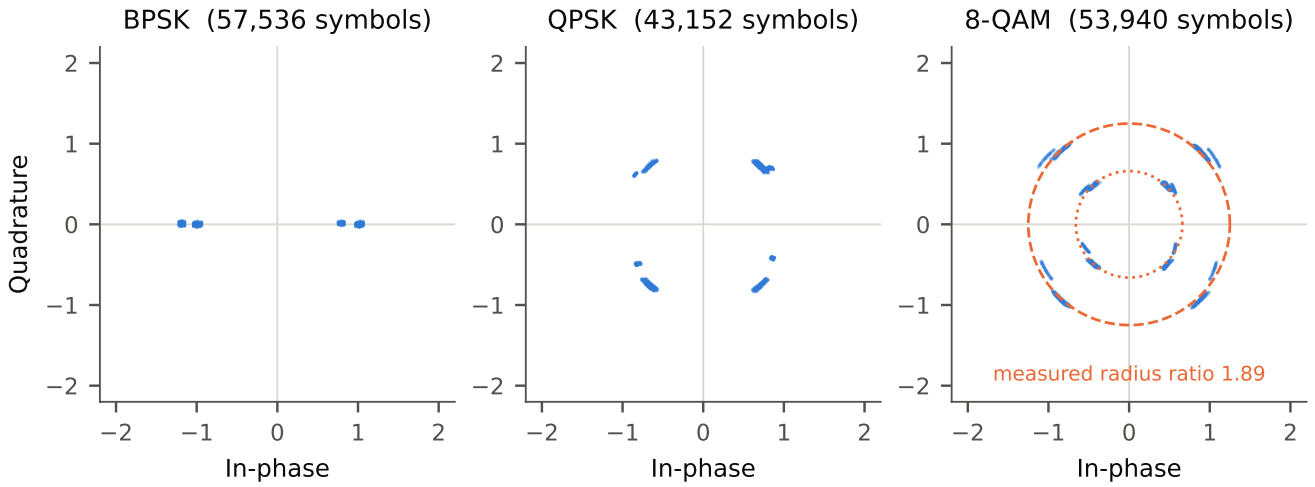


FIGURE 2. **Measured constellations, all records pooled across the five boards and normalised per record to unit RMS. The 8-QAM panel shows the two rings at the measured radius ratio of 1.89. No noise has been added.**

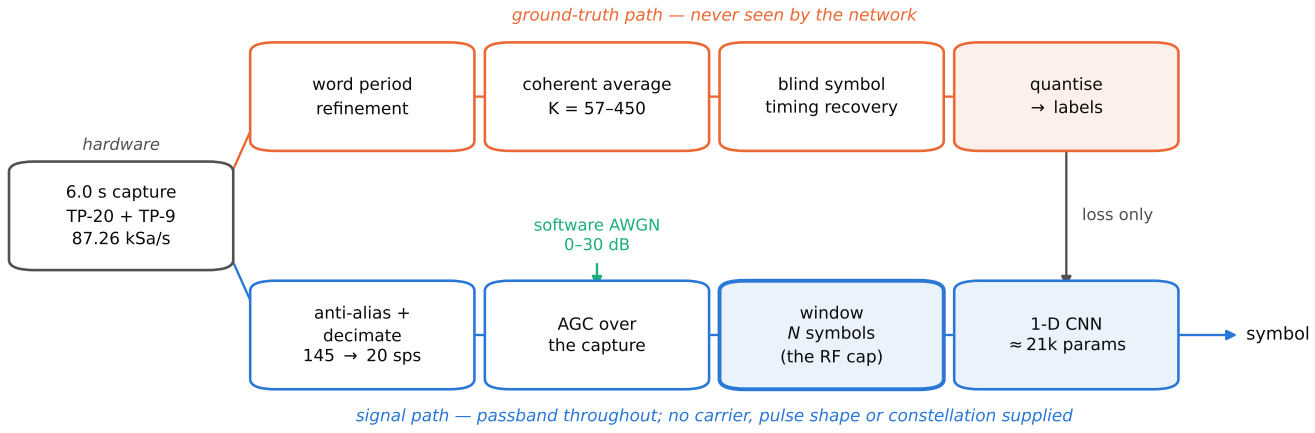


FIGURE 3. **Processing chain. The upper path produces ground-truth labels by coherent averaging; the lower path produces the network's input. The paths share only the raw capture, and the labels reach the network only through the loss.**

The period is refined by maximising the energy of the coherent average,

$$P^{\star} = \arg\max_{P} \ \mathrm{var}_m \left(t[m;P]\right), \tag{3}$$

searched over ± 6 samples in steps of 0.02 samples with M held fixed so that candidates are scored on a common grid. Writing $\sigma_r^2 = \mathrm{var}(x_k[m] - t[m])$ for the residual variance, the template signal-to-noise ratio is reported as

$$\mathrm{SNR}_t = 10\log_{10} \frac{\mathrm{var}(t) - \sigma_r^2/K}{\sigma_r^2}, \tag{4}$$

in which the second numerator term removes the averaging bias, so the quantity reported is the single-repetition signal-to-noise ratio rather than the flattered post-averaging one.

B. SYMBOL EXTRACTION

To prevent the final symbol from falling on the wrap boundary, the coherent template is first extended circularly prior to filtering. The subsequent downconversion is achieved without requiring an external carrier estimate; because the transmitted word contains exactly $2N_s$ carrier cycles for N_s symbols, the downconverted sequence is computed as:

$$b[n] = t_{\mathrm{ext}}[n]\, e^{-j2\pi(2N_s)n/M}, \tag{5}$$

where $b[n]$ represents the downconverted discrete-time signal sequence, $t_{ext}[n]$ is the circularly extended coherent template waveform, n is the discrete-time sample index,N_s denotes the total number of transmitted symbols within the data word, and M is the number of grid points per word established in Section IV-A. This formulation remains exact regardless of the underlying sample rate.

Following downconversion, the sequence $b[n]$ is passed through a matched filter. Symbols are then sampled at instants $(i+\tau)T_s$, where T_s is the symbol period and the fractional timing offset $\tau \in [0,1)$ is recovered blind by maximising the mean symbol energy:

$$\tau^\star = \arg\max_{\tau} \frac{1}{N_s} \sum_i \left|s_i(\tau)\right|^2. \tag{6}$$

where $s_i(\tau)$ denotes the complex output of the matched filter for the i-th symbol evaluated at the candidate timing offset τ, and i represents the discrete symbol index. Symbol-timing recovery is a prerequisite rather than a refinement. Extracting symbols without Equation (6) places decision instants part-way through transitions and collapses individual constellation radii—in our data, two of the eight 8-QAM radii fell to 0.20 against a nominal 1.31. This artefact is visually indistinguishable from a template wrap-boundary error and must be excluded before that diagnosis is reached.

Once timing is established, the residual constellation rotation is removed by an mth-power estimate:

$$\rho = \frac{1}{m} \arg\left(\mathbb{E}[s^m]\, e^{-jm\phi_{\mathrm{ref}}}\right), \qquad s' = s\, e^{-j\rho}, \tag{7}$$

where ρ is the estimated residual rotation angle, m is the modulation-dependent exponent ($m = 2$ for BPSK and $m = 4$ otherwise), and ϕ_{ref} is the target reference phase for the constellation. The variable s represents the optimally sampled complex symbol (corresponding to $s_i(\tau^*)$), s' is the resulting derotated symbol, and $\mathbb{E}[\cdot]$ denotes the sample mean taken over all ≈ 3600 symbols of the capture record. The rotation ϕ_{ref} must be applied inside the argument as written: for a reference phase of $45°$, the fourth-power sum lies near $180°$, which sits on the branch cut of the argument function. Subtracting the phase afterwards would erroneously place records whose phase is physically identical $90°$ apart. Furthermore, the expectation is calculated over the entire record rather than a single word, because a single word covering the constellation unevenly introduces enough bias to select an incorrect quadrant. Either of these errors would relabel every symbol of the affected record while leaving all per-record diagnostics deceptively intact.

C. VERIFICATION OF THE LABELS

Two independent checks confirm the recovered labels.

1) All sixteen BPSK records reproduce the board's 63-bit maximal-length sequence at 100%, against a chance level of 66.6% (95th percentile 69.8%) computed over 63 rotations and both polarities. The carrier is recovered blind by fourth-power estimation at 1199.915–1199.918 Hz across all five boards.
2) A single memoryless bijection from constellation point to dibit fits all three QPSK data words simultaneously at 100%—36 symbols explained by one four-point mapping. Across $20{,}000$ label permutations preserving the per-symbol occupancy counts, a perfect fit never occurred, giving $p < 5 \times 10^{-5}$.

The second check also resolves, for BPSK and QPSK, the mapping between transmitter configuration and transmitted symbol. For 8-QAM it cannot: a word covering all eight points exactly once admits a perfect bijection for any rotation, so the test is vacuous for a single word and remains unresolved.

TABLE 6. Receptive-field bound and model size. All configurations remain at or below the five-bit limit implied by (8).

Modulation	Symbols	Bits	Window	RF	Parameters
BPSK	5	5	100	93	20,932
QPSK	2	4	40	39	22,022
8-QAM	1	3	20	20	20,106

V. RECEIVER

A. THE RECEPTIVE-FIELD BOUND

The transmitted sequence is a maximal-length sequence of linear order six, satisfying:

$$s[i+6] = s[i] \oplus s[i+5], \tag{8}$$

where $s[i]$ represents the binary value of the i-th bit in the transmitted sequence, i is the discrete bit index, and $\oplus$ denotes modulo-2 addition (exclusive OR). Because of this linear recursion, any six consecutive bits determine the seventh exactly. No five consecutive bits do: each five-bit window occurs twice per period with distinct successors. [49].

Consequently a receiver whose input contains at most five bits cannot predict a symbol from its neighbours, and must extract it from the waveform. We enforce this through the *input* rather than through the layer stack: the network receives a window of exactly N symbols and labels the symbol within it, and cannot attend to a sixth bit because no sixth bit is present in its input. The formulation is verifiable by inspection and is unaffected by later changes to the architecture. Table 6 gives the resulting configuration.

For a stack of one-dimensional convolutions, the receptive field RF in input samples is calculated as:

$$\mathrm{RF} = 1 + \sum_l (k_l - 1) \prod_{j<l} s_j, \tag{9}$$

where k_l is the kernel size of the l-th convolutional layer, s_j represents the stride of the j-th layer, and the product is taken over all layers j preceding layer l. This calculated RF is asserted against the window width in code. The bound is tightest for 8-QAM, where a single symbol already carries three bits; this is the configuration in which the argument is weakest, and accordingly the one for which we provide the largest number of independent held-out words.

B. SIGNAL CONDITIONING

Waveforms are low-pass filtered with a zero-phase FIR and decimated from 145.44 to 20 samples per symbol, and are presented in *passband*. Conversion to baseband is not performed, as it would amount to supplying the carrier frequency.

Amplitude is normalised by an automatic gain control computed over the whole record:

$$\tilde{x}[n] = x[n] \Big/ \sqrt{\tfrac{1}{L} \textstyle\sum_l x[l]^2}, \tag{10}$$

where $\tilde{x}[n]$ is the amplitude-normalised waveform sample at discrete-time index n, $x[n]$ is the filtered and decimated passband waveform sample, L is the total number of samples in the entire capture record, and l is the discrete summation index. This record-level normalisation removes device gain—one board's output is 40% below the others—and attenuator setting, while preserving symbol-to-symbol amplitude ratios. Per-window normalisation must not be used. An 8-QAM window is a single symbol, and forcing it to unit energy erases the inner/outer ring distinction, which carries one of the three bits per symbol. In our measurements, this held the 8-QAM symbol error rate at 0.50 with no improvement from 0 dB to noiseless, because the ring bit was pure guesswork. Additive white Gaussian noise is applied during training with the signal-to-noise ratio drawn uniformly from 0 to 30 dB. The noise power is referenced to the record level of Equation (10) so that all symbols in a record experience the same noise power. Referencing to the individual window would give inner-ring symbols less noise than outer-ring symbols, which no physical channel does.

C. ARCHITECTURE AND OBJECTIVE

The receiver is a four-layer one-dimensional convolutional network with 32 channels, a full-width learned combiner in place of global pooling, and two output heads: symbol-class logits and a latent $(\hat{I}, \hat{Q})$ pair. Global average pooling was deliberately avoided: it makes the effective receptive field equal to the whole input while a kernel-based calculation reports a smaller figure, so the number one would quote would not be the operative one.

The training objective is:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda \min_{c \in \mathcal{C}} \left\| (\hat{I}, \hat{Q}) - c \right\|^2, \tag{11}$$

where $\mathcal{L}$ is the total training loss, $\mathcal{L}_{CE}$ is the cross-entropy loss derived from the symbol-class logits, and λ is a scalar hyperparameter determining the weight of the constellation-geometry penalty. The term $(\hat{I}, \hat{Q})$ represents the continuous two-dimensional coordinates produced by the latent output head, $\mathcal{C}$ is the measured constellation set from Fig. 2(comprising two antipodal points for BPSK, a single ring of four for QPSK, and a two-ring star of radius ratio 1.89 for 8-QAM), c is a candidate constellation point within $\mathcal{C}$, and $||\cdot||^2$ denotes the squared Euclidean distance.

The cross-entropy term $\mathcal{L}_{CE}$ is class-weighted to compensate for the intrinsic imbalance of the transmitted words.

TABLE 7. Symbol error rate, learned (L) and blind classical (C) receivers, leave-one-pattern-out, mean over folds.

	BPSK		QPSK		8-QAM	
SNR	L	C	L	C	L	C
0 dB	0.0001	0.0000	0.0093	0.0037	0.1319	0.1219
3 dB	0.0000	0.0000	0.0011	0.0003	0.0432	0.0381
6 dB	0.0000	0.0000	0.0000	0.0000	0.0065	0.0058
9 dB	0.0000	0.0000	0.0000	0.0000	0.0003	0.0003

TABLE 8. 8-QAM symbol error rate by evaluation split, mean over folds.

Split	Folds	0 dB	3 dB	6 dB	9 dB
Random 80/20	1	0.1269	0.0387	0.0054	0.0005
Leave-one-pattern-out	5	0.1319	0.0432	0.0065	0.0003
Leave-one-board-out	5	0.1327	0.0451	0.0105	0.0021
Leave-one-atten.-out	2	0.1367	0.0452	0.0071	0.0004

Optimisation uses Adam at 2×10^{-3} with cosine decay over 25 epochs and a batch size of 512.

D. EVALUATION SPLITS

Partitions are constructed at *record* level. Symbols within one record are repetitions of a single data word through a single signal path, so a symbol-level partition would place near-duplicates on both sides. Four splits are reported: leave-one-pattern-out, leave-one-board-out, leave-one-attenuation-out, and a random 80/20 partition, the last included explicitly as the optimistic reference. Thirty-five folds are evaluated in total.

VI. RESULTS

A. ERROR RATE AGAINST A BLIND CLASSICAL RECEIVER

The reference receiver performs fourth-power carrier recovery, matched filtering and hard decision, and is given no parameter the network is not given. Noise is identical for both. Fig. 4 and Table 7 report leave-one-pattern-out performance.

The two receivers are equivalent. The classical receiver retains a small advantage at 0 dB—0.1219 against 0.1319 for 8-QAM, a relative difference of 8%—which is expected, as a matched filter is near-optimal for additive Gaussian noise. Above 9 dB the learned receiver reaches zero errors while the classical receiver retains a residual floor of 2×10^{-4}. We interpret neither difference as a performance claim.

B. GENERALISATION

Holding out an entire data word costs essentially nothing. At 0 dB the leave-one-pattern-out and random splits differ by 0.005 in symbol error rate, within run-to-run variation (Fig. 5). This is the empirical counterpart of the bound of Section V-A: a model exploiting sequence structure would score markedly better on a random split, in which training and test partitions share the same data word. It does not.

Cross-device transfer is measurably harder. Leave-one-board-out degrades to 0.0021 at 9 dB against 0.0003 for leave-one-pattern-out, a factor of seven. The effect is small

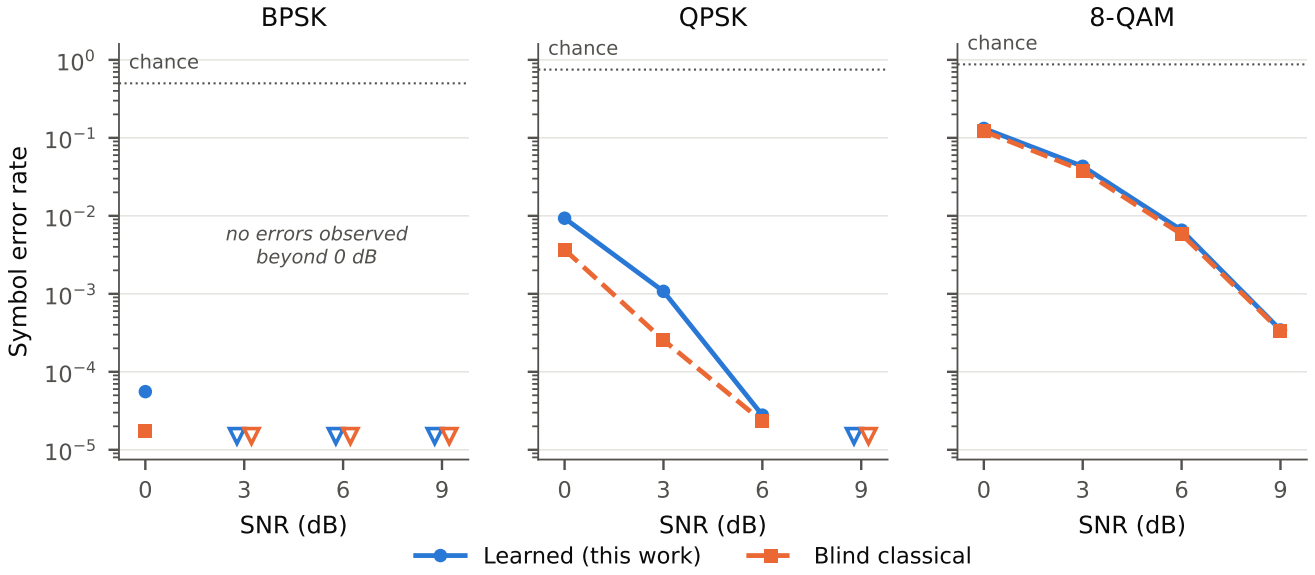


FIGURE 4. Symbol error rate against signal-to-noise ratio for the learned and blind classical receivers, leave-one-pattern-out. Hollow triangles mark operating points at which no errors occurred in the test set, drawn at the $1/N$ resolution floor. All quantities saturate above 9 dB.

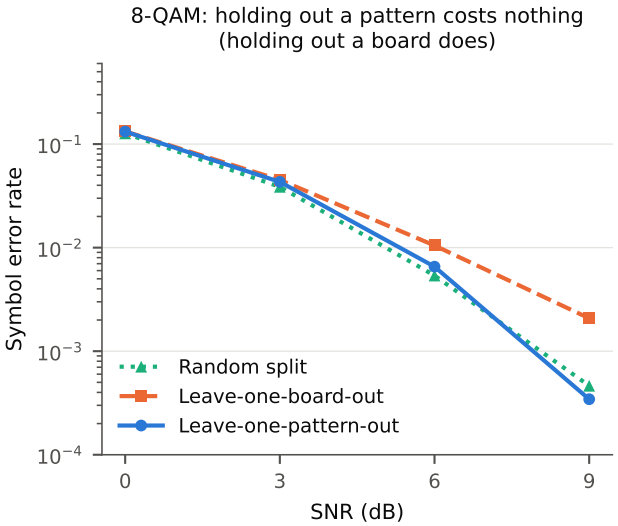


FIGURE 5. 8-QAM symbol error rate by evaluation split. Withholding a complete data word is indistinguishable from a random partition; withholding a board is not.

in absolute terms but consistent across folds, and is the only split exhibiting a systematic gap.

Per-fold results are given in Table 9. Two folds are thin and are reported with their training size: BPSK with the pseudorandom sequence held out trains on a single record and QPSK with its primary word held out trains on two. The 8-QAM folds, where the bound of Section V-A is tightest, are the best conditioned: four of five train on fourteen records.

C. NEGATIVE CONTROLS

Permuting labels among the symbols of each fold leaves performance at chance, confirming that no path exists from input to target other than the waveform.

The second control requires a qualification that is easy to overlook. Labels are a periodic tiling of the transmitted word, so a symbol's index within that word determines its label exactly. Across all 148 pairs of records carrying the same word—including pairs drawn from different boards and different attenuations—the label sequences are identical up to a cyclic shift. A predictor reading only the symbol index therefore attains a symbol error rate of zero on any split whose test partition reuses a training word, which is the case for every leave-one-board-out, leave-one-attenuation-out and random fold. On those splits the position-only figure is not a control and we do not report it.

It is a control on leave-one-pattern-out, where the withheld word is absent from training and is not a cyclic shift of any training word: the best cyclic agreement between distinct words is 0.500 for QPSK and 0.375 for 8-QAM, against chance agreements of 0.250 and 0.125. There the lookup sits at chance for all three modulations—0.5147 against 0.500, 0.7862 against 0.750, and 0.8523 against 0.875 (Table 10, Fig. 6).

This concerns the control, not the receiver. The network is given a window of waveform samples and nothing else: it receives no symbol index and classifies each window indepen-

TABLE 9. Per-fold results. *Train rec.* is the number of records in the training partition; the two folds with fewer than three are the thin folds discussed in the text. *Position-only* is a lookup from a symbol's index within the data word to its most frequent label, against chance levels of 0.500, 0.750 and 0.875. It is a meaningful control only where the withheld word is absent from training; every leave-one-board-out fold reuses the training word, position then determines the label exactly, and the entry is marked *n/a* (Section VI-C).

Modulation	Held out	Train records	Test symbols	SER at 0 dB	SER at 6 dB	Position-only
Leave-one-pattern-out						
BPSK	PRBS63	1	53,880	0.0001	0.0000	0.5290
BPSK	S1	15	3,592	0.0000	0.0000	0.5003
QPSK	Q1	2	35,950	0.0174	0.0001	0.7751
QPSK	Q2	11	3,595	0.0042	0.0000	0.7502
QPSK	Q3	11	3,595	0.0064	0.0000	0.8334
8-QAM	T1+T2	4	39,556	0.1519	0.0119	0.8864
8-QAM	T3	14	3,596	0.1310	0.0072	0.8749
8-QAM	T4	14	3,596	0.1313	0.0028	0.7500
8-QAM	T5	14	3,596	0.1271	0.0044	0.8751
8-QAM	T6	14	3,596	0.1185	0.0064	0.8751
Leave-one-board-out						
BPSK	B1	12	14,368	0.0001	0.0000	n/a
BPSK	B2	13	10,776	0.0000	0.0000	n/a
BPSK	B3	13	10,776	0.0002	0.0000	n/a
BPSK	B4	13	10,776	0.0000	0.0000	n/a
BPSK	B5	13	10,776	0.0002	0.0000	n/a
QPSK	B1	8	14,380	0.0070	0.0000	n/a
QPSK	B2	10	7,190	0.0054	0.0000	n/a
QPSK	B3	10	7,190	0.0049	0.0000	n/a
QPSK	B4	10	7,190	0.0036	0.0000	n/a
QPSK	B5	10	7,190	0.0174	0.0001	n/a
8-QAM	B1	8	25,172	0.1287	0.0057	n/a
8-QAM	B2	13	7,192	0.1194	0.0049	n/a
8-QAM	B3	13	7,192	0.1193	0.0057	n/a
8-QAM	B4	13	7,192	0.1283	0.0060	n/a
8-QAM	B5	13	7,192	0.1675	0.0302	n/a

TABLE 10. Controls, leave-one-pattern-out folds. Both must fail, and both do. The position-only control is restricted to this split for the reason given in the text.

Modulation	Chance	Shuffled	Position-only	Model
BPSK	0.500	0.4979	0.5147	0.0000
QPSK	0.750	0.7349	0.7862	0.0000
8-QAM	0.875	0.8834	0.8523	0.0065

TABLE 11. Ablation of the constellation term of Equation(11), leave-one-board-out. Differences lie within run-to-run variation.

	8-QAM			QPSK
λ	0 dB	3 dB	6 dB	0 dB
0.0	0.1246	0.0370	0.0060	0.0058
0.2	0.1261	0.0359	0.0056	0.0051
1.0	0.1276	0.0364	0.0058	0.0054

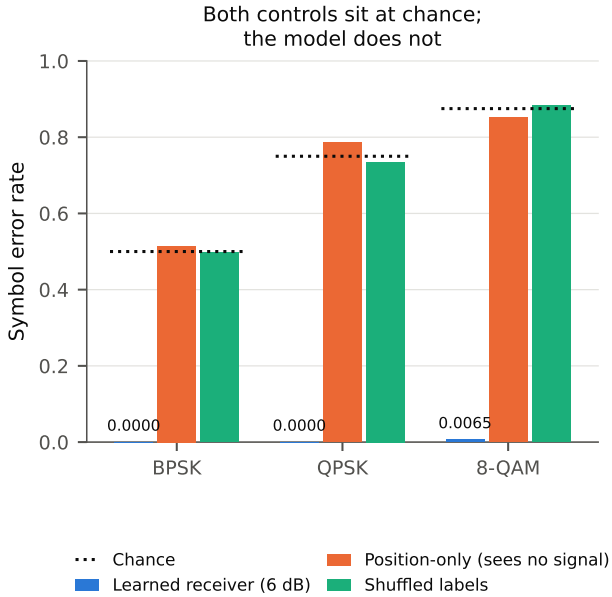


FIGURE 6. Both negative controls remain at their respective chance levels while the receiver does not. Leave-one-pattern-out folds.

dently, so on no split can it exploit the positional redundancy that defeats the lookup.

D. RECOVERED CONSTELLATION GEOMETRY

The constellation term of equation (11) was ablated across two decades of λ (Table 11). *The constellation prior does not improve error rate.* Its role is interpretability: it aligns the latent $(\hat{I}, \hat{Q})$ head with the physical constellation. We do not present it as a performance mechanism.

The geometry is nevertheless recovered, by the classifier alone and from data. Examining the destination of errors at 0 dB (Fig. 7), 79.5% of 8-QAM errors land on the geometrically nearest constellation point against a chance level of 14.3%, an enrichment of 5.6×; for QPSK the correspond-

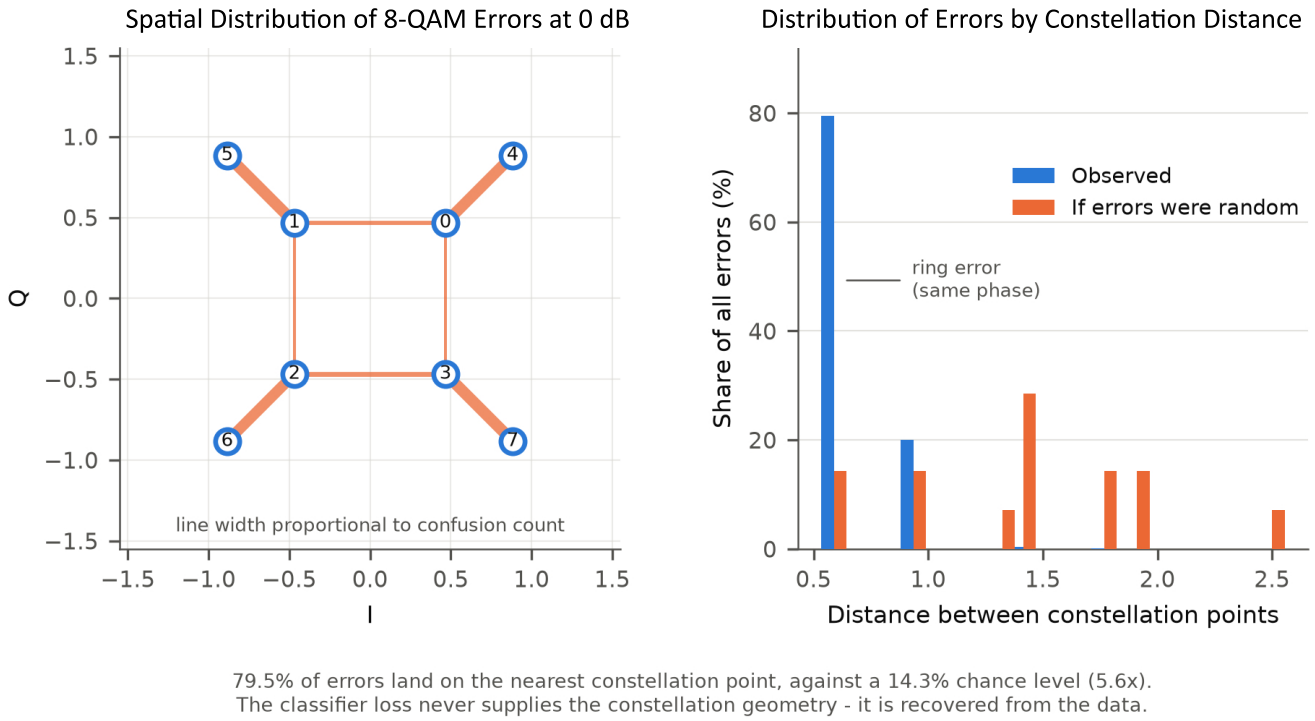


FIGURE 7. Destination of 8-QAM errors at 0 dB. Left: confusions drawn on the constellation, line width proportional to count. Right: share of errors against inter-point distance, compared with a uniform-error null.

ing figures are 100% against 66.7%. Of the 56 off-diagonal cells of the confusion matrix, 36 are exactly zero, and the populated cells are the same-phase ring transition and the adjacent-quadrant transition—the two shortest distances in the constellation.

A receiver implementing a memorised waveform-to-label mapping would distribute errors uniformly over incorrect classes. This one confuses each point with its metric neighbours, which is where additive noise transports a symbol.

E. EMPIRICAL PERFORMANCE OF THE ON-BOARD DEMODULATORS

The captures also characterise the boards' own demodulators, measured against the known transmitted sequence rather than by self-consistency (Fig. 8). Across all five units, agreement with the transmitted sequence is at chance for every record below 509 mV output level and reaches 92–100% for every record above 626 mV. A single threshold within that interval accounts for all five boards.

Board-level behaviour differs in kind rather than degree. One board's BPSK path delivers 391 mV where the others deliver 626–663 mV and never reaches threshold, while its QPSK and 8-QAM outputs are indistinguishable from the others (cross-board waveform correlation $+0.9987$ and $+0.9991$, against $+0.9769$ for BPSK). A second board crosses the threshold but locks imperfectly, at 92.1%.

We recommend against self-consistency as a lock criterion. A demodulator that has lost lock also repeats. On one board at 329 mV, self-consistency read 0.979 while agreement with the transmitted sequence was at chance: the unlocked demodulator had entered a stable period-two state, inverting its output every word period (autocorrelation -0.93 at one word period, $+0.90$ at two). Four records in this study would have been misclassified as locked by that metric.

Finally, the boards' own QPSK demodulator failed to lock on the primary QPSK word on all five units, at both attenuator settings, although the fingerprint check confirms the word was correctly loaded. No hardware baseline is therefore available for QPSK.

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VII. DISCUSSION

The principal result of this work is methodological. When a learned receiver is trained on a periodic transmitted sequence, held-out evaluation alone does not establish that the receiver

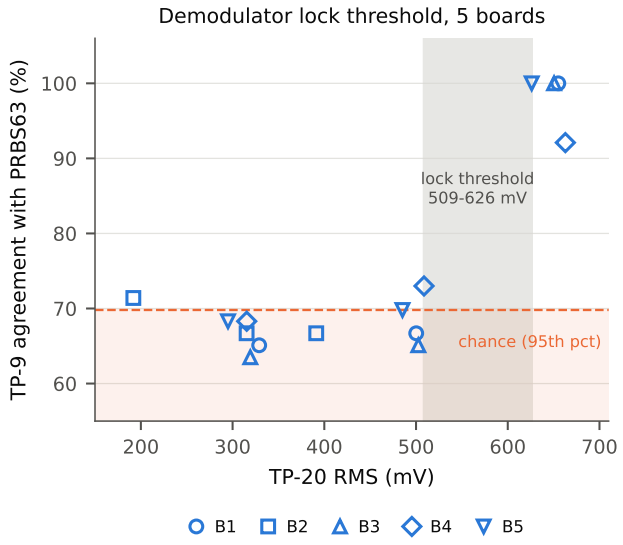


FIGURE 8. Demodulator lock threshold across five boards, measured against the known transmitted sequence. A single threshold accounts for every unit.

demodulates. The property can instead be guaranteed: by bounding the receiver's input below the linear order of the transmitted sequence, prediction-from-context becomes impossible rather than unlikely, and the guarantee survives any change to model capacity or training procedure.

That guarantee has an empirical signature—no gap between a random partition and a leave-one-pattern-out partition—which is then available as a consistency check rather than as the primary evidence. We observe that signature (Section VI-B). The two lines of argument are independent: the bound would hold even had the gap been large, and the small gap would be suggestive but not conclusive without the bound.

We note that the bound is cheap to adopt. It requires knowing the linear order of the transmitted sequence, which is a property of the test signal rather than of the receiver, and constraining the input window accordingly. Where a test sequence has high linear complexity the constraint is loose; where it is a short m-sequence, as in much laboratory practice, it is tight and necessary.

The parity result of Section VI-A should be read in this light. A matched-filter receiver with correct carrier recovery is near-optimal for the channel studied here, and a learned receiver cannot be expected to exceed it. The substantive observation is that parity is reached without supplying the carrier frequency, the pulse shape, the constellation geometry or the decision boundaries, and that the constellation is recovered from data with a $5.6\times$ enrichment toward nearest-neighbour errors.

The one systematic gap we observe is cross-device: leave-one-board-out is consistently worse than leave-one-pattern-out at high signal-to-noise ratio. This is the expected direction the boards differ measurably in output level, constellation tightness and transmit-filter delay and it suggests that device-invariant training, rather than sequence generalisation, is where the remaining headroom lies for this class of receiver.

Beyond the evaluation of the learned receiver itself, the empirical performance of the hardware baseline in Section VI-E is of independent interest to anyone evaluating aged laboratory equipment. That a demodulator which has lost lock can present a perfectly self-consistent output is not, in our experience, widely appreciated, and any protocol treating repeatability as evidence of correct operation is exposed to it. The failure is also invisible on an oscilloscope: an unlocked digital output looks entirely normal at any time base. We therefore recommend that lock be established only by comparison against a known transmitted sequence.

VIII. LIMITATIONS

Content diversity. The corpus is large in noise realisations and hardware variation but small in distinct transmitted content: 8-QAM comprises five independent 24-bit words, or 40 distinct symbol instances.

Device count. Five functional units support a five-fold cross-device evaluation. The device-to-device differences we observe are large enough to be measured but the sample is too small to characterise their distribution.

Synthetic noise. The hardware supplies the waveform carrier, filter response, quantisation, device-to-device variation while additive white Gaussian noise is applied in software. The boards' own noise control contributes approximately $+0.02$ dB within the signal band and is unusable as a signal-to-noise axis.

Symbol timing is supplied. The receiver is given neither the carrier frequency nor the pulse shape, but its input window is symbol-synchronous, which supplies the symbol rate and the sampling phase. This is not incidental: under a one-symbol receptive field for 8-QAM, blind timing recovery is impossible, so the receptive-field bound and timing-blindness cannot both hold. The system should be described as operating on symbol-synchronous windows from a conventional timing front-end [48].

Carrier sensitivity does not support a robustness claim. We measured whether a classical receiver's dependence on carrier knowledge constitutes a practical disadvantage. Estimating the carrier blind from eight symbols (13.3 ms) gives an error of 1.02 Hz, yet demodulating those same eight symbols with that estimate yields zero errors, because over 13.3 ms a 1 Hz offset corresponds to 4.9° of phase rotation. Estimate error and tolerance contract together, and the classical receiver was self-consistent at every observation length tested, from 8 to 3,596 symbols. Reporting a collapse under an externally imposed carrier offset would misrepresent a receiver that does not make that error, and we therefore do not report one.

Operating regime. A single carrier frequency, a single symbol rate, and an additive Gaussian channel without multipath or fading. Generalisation beyond this regime is not evidenced here.

IX. CONCLUSION

We have presented a hardware dataset of 43 dual-channel captures from five distinct communication trainer boards, together with a convolutional receiver whose input is bounded below the linear order of the transmitted sequence, so that sequence memorisation is excluded by construction rather than argued from held-out performance. The receiver attains symbol error rates indistinguishable from a blind matched-filter receiver across BPSK, QPSK and star 8-QAM while receiving none of the parameters that receiver requires, and its errors concentrate on nearest constellation neighbours at $5.6\times$ the chance rate, indicating that the constellation geometry is recovered from the waveform. Two negative results are reported: a constellation-geometry loss term does not improve error rate, and a classical receiver's dependence on carrier knowledge did not, under measurement, constitute a practical disadvantage.

Future work should address the two limitations that bound the present study most tightly: the number of distinct transmitted words, which sets the resolution of the leave-one-pattern-out evaluation, and the number of devices, which sets the resolution of the cross-device evaluation.

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